

EXPLORATORY DATA ANALYSIS REPORT

Project 2 — Uncovering the Story in the Data

DecodeLabsIndustrialTraining · Batch2026 · 2026

Problem Statement	DecodeLabs has 1,200 e-commerce orders spanning Jan 2023 – Jun 2025 across 7 product categories. The business needs to understand: Which products, channels, and payment methods drive the most revenue? Where is revenue being lost? What patterns and outliers exist in order behaviour?
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DATASET SNAPSHOT

Records	Columns	Date Range	Total Revenue	Avg Order Value
1,200	14	Jan 2023 – Jun 2025	Rs 12,64,762	Rs 1,053.97

DESCRIPTIVE STATISTICS

Metric	Quantity	Unit Price (Rs)	Total Price (Rs)	Items in Cart
Count	1,200	1,200	1,200	1,200
Mean	2.95	356.41	1,053.97	5.49
Median	3.00	364.21	823.62	5.00
Std Dev	1.41	197.18	819.86	2.28
Min	1	11.39	11.39	1
Q1	2	186.06	410.52	4
Q3	4	521.57	1,578.48	7
Max	5	699.93	3,456.40	10
Skewness	0.005	-0.019	0.784	0.013

Note: TotalPrice skewness = 0.784 indicates a mild right skew — a small number of high-value orders pull the mean above the median. Median (Rs 823.62) is the more reliable measure of typical order value.

OUTLIER DETECTION — IQR Method

Column	Lower Fence	Upper Fence	Outliers Found	Classification
Quantity	-1.0	7.0	0	Clean ✓
UnitPrice	-316.70	1,024.35	0	Clean ✓
TotalPrice	-1,341.41	3,330.41	8	OUTLIERS FOUND
ItemsInCart	-0.5	11.5	0	Clean ✓

8 TotalPrice outliers identified (values above Rs 3,330.41). These are high-value legitimate orders — flagged for VIP customer investigation, not deletion.

REVENUE ANALYSIS

Revenue by Product Category

Product	Total Revenue (Rs)	Avg Order (Rs)	Orders	Revenue Share
Chair	1,95,620	1,098.99	178	15.5% ▲
Printer	1,95,613	1,080.73	181	15.5% ▲
Laptop	1,92,127	1,110.56	173	15.2%
Tablet	1,86,569	1,042.28	179	14.8%
Monitor	1,75,651	1,077.62	163	13.9%
Desk	1,67,460	985.06	170	13.2%
Phone	1,51,722	972.58	156	12.0% ▼

Revenue by Referral Channel & Payment Method

Channel	Revenue (Rs)	Orders	Payment	Revenue (Rs)	Avg (Rs)
Instagram	2,75,285	259	Credit Card	2,63,848	1,127.55
Email	2,61,809	250	Online	2,62,443	1,017.22
Google	2,50,441	241	Cash	2,59,786	1,056.04
Facebook	2,50,411	228	Gift Card	2,46,324	1,070.97
Referral	2,26,816	222	Debit Card	2,32,361	1,001.56

ORDER STATUS DISTRIBUTION

Status	Count	% of Total	Revenue Impact
Cancelled	250	20.8%	Revenue Lost
Returned	247	20.6%	Revenue Lost
Pending	237	19.8%	At Risk
Shipped	235	19.6%	Active
Delivered	231	19.3%	Fulfilled

Critical Finding: 41.4% of all orders are Cancelled or Returned — this represents significant revenue leakage and requires urgent business attention.

CORRELATION ANALYSIS

Variable	Quantity	UnitPrice	TotalPrice	ItemsInCart
Quantity	1.000	0.015	0.615	0.650
UnitPrice	0.015	1.000	0.717	0.001
TotalPrice	0.615	0.717	1.000	0.393
ItemsInCart	0.650	0.001	0.393	1.000

Dark Green = Strong ($r \geq 0.60$)

White = Weak ($r < 0.40$)

Key: UnitPrice is the strongest predictor of TotalPrice ($r=0.717$). Quantity and ItemsInCart also show meaningful positive correlations with revenue.

KEY INSIGHTS & RECOMMENDATIONS

CRITICAL	Order Fulfilment Crisis Finding: 41.4% of orders are Cancelled (250) or Returned (247). This is the single largest revenue risk in the dataset. Recommendation: Audit cancellation/return reasons. Introduce post-order feedback forms. Improve product descriptions and images.
HIGH	Top Revenue: Chair & Printer Lead Finding: Chair (Rs1,95,620) and Printer (Rs1,95,613) each contribute ~15.5% of total revenue, significantly ahead of Phone (12%). Recommendation: Prioritise inventory and marketing for top-performing categories. Consider bundling Chair with Desk.
HIGH	Instagram is the #1 Revenue Channel Finding: Instagram drives Rs 2,75,285 in revenue (259 orders) — the highest of all referral sources. Recommendation: Increase Instagram ad spend and influencer partnerships. Replicate successful content formats.
MEDIUM	Credit Card = Highest Avg Order Value Finding: Credit Card users spend Rs1,127.55 on average — 6.9% above the overall mean of Rs1,053.97. Recommendation: Offer credit card exclusive deals or EMI options to encourage higher basket sizes.
MEDIUM	Price Drives Revenue ($r=0.717$) Finding: UnitPrice has the strongest correlation with TotalPrice ($r=0.717$). Premium products are the primary revenue engine. Recommendation: Invest in premium product line expansion. High price, lower quantity orders yield better per-order revenue.

LOW

Mid-Year Revenue Peaks

Finding: June 2024 (Rs 68,069) and May 2023 (Rs 63,837) show consistent mid-year revenue spikes. **Recommendation:** Launch seasonal promotions in April-May each year to amplify the existing mid-year demand trend.